ELEC5305 Project Progress Description

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Project	Investigating Phase Limitations in STFT-Based Speech Enhancement Using Oracle-Phase Analysis

Project Description

Following the feedback on my original proposal, I refined the project to focus specifically on the limitations of noisy-phase reuse in STFT-based speech enhancement. I implemented controlled VoiceBank + DEMAND experiments using spectral subtraction and Wiener filtering, comparing noisy-phase reconstruction with clean oracle-phase reconstruction. I also added the suggested perfect-magnitude diagnostic, phase-disagreement analysis, and STOI, PESQ and SI-SDR evaluation.

The full classical experiment now covers **360 controlled mixtures** across three noise types and three SNR levels. I also incorporated a pretrained **MP-SENet** as the suggested modern phase-aware comparison and performed a matched comparison on **167 successfully processed mixtures**. Representative spectrograms, objective results and listening samples have also been added.

The current results show that noisy-phase reuse is a measurable limitation for classical enhancement, but its effect is metric-dependent. The perfect-magnitude experiment also shows that magnitude-estimation accuracy remains a major limiting factor, while MP-SENet substantially outperforms the classical methods on the matched subset.

GitHub Project Site

https://sliu2040977-hash.github.io/elec5305-project-540569223/

GitHub Repository

https://github.com/sliu2040977-hash/elec5305-project-540569223